INFORMATION RETRIEVAL

Information Retrieval: Searching in the 21st Century Edited by A. Göker & J. Davies
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INFORMATION RETRIEVAL: SEARCHING IN THE 21ST CENTURY

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Dedication

I would like to thank my husband, parents and wider family for all their support to my efforts to reach my full potential. This book is one of the tangible outcomes of this process. My father, in particular, would have loved to have seen it.

Ayşe Göker

Contents

Foreword

In the forty years since I started working in the field, and indeed for some years before that (almost since Calvin Mooers coined the term *information storage and retrieval* in the 1950s), there have been a significant number of books on information retrieval. Even if we ignore the more specialist research monographs and the 'readers' of previously published papers, I can find on my shelves or in my mental library many books that attempt (probably with the IR student in mind) to construct a coherent and systematic way of defining and presenting information retrieval as a field of study and of application.

Often such a book is the work of a single author, or perhaps a pair working together. Such works can clearly have an advantage in respect of coherence; the field is necessarily presented from a single viewpoint. On the other hand, they can also suffer for the same reason. The IR field is rich (more so now than it has ever been), and it is difficult within a single viewpoint to do justice to this richness. Readers, on the other hand, have to be constructed out of the materials to hand: the published papers, each of which has taken its own view, probably with a much narrower field of vision, and different from that of the other chosen papers.

The present book attempts the tricky task of combining the breadth of vision of multiple authors with the coherence of a single integrated work. The richness of the field is apparent in the range of chapters: from formal mathematical modelling to user context, from parallel computation to semantic search.

The topics covered also vary greatly in their historical association with the field. *Categorisation*, for example, has been around as an IR technique for quite a long time – though Stuart Watt brings a new perspective. *Mobile search* (David Mountain, Hans Myrhaug and Ayşe Göker), however, is a relatively recent development. The use of formal models (*information retrieval models*, Djoerd Hiemstra) goes back almost to the beginning, as does experimental evaluation (*user-centred evaluation of information retrieval systems*, Pia Borlund), though in both cases there have been huge changes in the past decade.

This same decade has witnessed the huge growth of the World Wide Web, and the developing dominance of web search engines (*web information retrieval*, Nick Craswell and David Hawking) as the glue which holds the web together. For many people today, IR *is* web search. It is true that there has been a huge amount of influence in both directions: search engines are largely based on techniques from both the IR research community and from previous operational systems, while IR research and practice in other environments has learnt a great deal from the forcing-house that is the web search space. This dominance of the web as *the* domain of interest is well reflected in many of the chapters in the present volume.

It is important, however, to remember that IR is not all about web search, and that the web space presents both problems and opportunities which differ from those in other domains. The desktop, the enterprise, specialist collections such as scientific papers are all examples of different domains for which search functionality is a fundamental requirement. There are references to several of these throughout the book, but specific domains with their own chapters are *multimedia resource discovery* (Stefan Rüger) and *image users' needs and searching behaviour* (Stina Westman). The user theme is taken further in the *context and information retrieval* (Ayşe Göker, Hans Myrhaug and Ralf Bierig). More generic problem areas are addressed in *cross-language information retrieval* (Daqing He and Jianqiang Wang), in *semantic search* (John Davies, Alistair Duke and Atanas Kiryakov) and in the chapter on *natural language processing* (Tony Russell-Rose and Mark Stevenson). Finally, a chapter on *performance issues and parallelism* (Andrew MacFarlane) addresses more technical computing concerns.

Information retrieval, from being the rather arcane subject in which I did my masters degree forty years ago, has become one of the defining technologies of the twenty first century. I believe the present book does justice to this status.

Stephen Robertson
2008

Preface

This project originated during Ayşe Göker's time as Chair of the British Computer Society Information Retrieval Specialist Group (BCS IRSG) when John Davies presented the initial opportunity. The IRSG Committee took this up and as editors we broadened the contributors list.

Each project has its challenges and this book is no exception. We thank Andrew MacFarlane of City University London first and foremost for his collegiate support and superb organisational skills combined with subject knowledge. He gave us a great boost, particularly in the latter part of the project.

We would also like to thank Margaret Graham who provided initial input on the project.

We would like to thank colleagues at City University London who each contributed in significant ways: David Bawden, Jonathan Raper, Jo Wood, Jason Dykes, and Tamara Eisenschitz. Likewise, Stuart Watt, previously of Robert Gordon University and now a consultant. Last but not least, we thank Hans Myrhaug of AmbieSense for his wholehearted commitment and support. He has been willing to carry out any task required to move the project forward. We also appreciate his fantastic eye for detail, and his skills in graphic design.

Each chapter was reviewed by at least four referees looking at the subject matter, coverage, and readability. We thank them accordingly. The referees were as follows:

Leif Azzopardi, University of Glasgow; Ralf Bierig, Rutgers University; Malcolm Clark, Robert Gordon University; John Davies, BT Research; Tamara Eisenschitz, City University London; Ayşe Göker, City University London; Sarah Hinton, Wiley; Fang Huang, Robert Gordon University; Rowan January, Wiley; Paul Lewis, Southampton University; Andrew MacFarlane, City University London; Carol Peters, Italian National Research Council; Stefan Rüger, The Open University; Ian Ruthven, Strathclyde University; Tamar Sadah, Ex Libris Ltd; Aidan Slingsby, City University London; Alan Smeaton, Dublin City University; Sarah Tilley, Wiley; Fiona Walsh, Robert Gordon University.

About the Editors

Ayşe Göker

Dr. Ayşe Göker is a senior academic at City University London. Her research since the early 90s has focused on developing novel search techniques and environments, with an emphasis on personalised and context-sensitive information retrieval and management systems. These occur particularly within mobile and wireless computing, and also in bibliographic and web environments. Her skills are in identifying user needs and developing innovative systems that meet them. In international collaborations she has also been successful, with extensive experience in designing projects and managing teams to implement them. On the teaching side, Ayşe has developed course modules in information systems on several degree programmes at both postgraduate and undergraduate levels.

Ayşe is also a company co-founder of AmbieSense Ltd, a mobile information system company. This project began as the AmbieSense EU-IST project at Robert Gordon University, Aberdeen where she was a Reader and project leader. She holds a lifetime Enterprise Fellowship from the Royal Society of Edinburgh and Scottish Enterprise. More recently she was selected for and completed the Massachussetts Institute of Technology (MIT) Entrepreneurship Development Program in Boston, USA.

In her profession, she has been the Chair of the British Computing Society's Specialist Group in Information Retrieval (BCS IRSG) (2000-2005). She was recognised for the totality of her endeavours by becoming a finalist in the Blackberry Women & Technology Awards (2005) for Best Woman in Technology (Academia).

John Davies

Dr John Davies leads the Semantic Technology research group at BT. Current interests centre around the application of semantic web technology to business intelligence, information integration, knowledge management and service-oriented environments. He is Project Director of the €12m ACTIVE EU integrated project. He co-founded the European Semantic Web conference series. He is also chairman of the European Semantic Technology Conference and a Vice-President of the Semantic Technology Institute. He chairs the NESSI Semantic Technology working group. He has written and edited many papers and books in the areas of the semantic technology, web-based information management and knowledge management; and has served on the program committee of numerous conferences in these and related areas. He is a Fellow of the British Computer Society and a Chartered Engineer. Earlier research at BT led to the development of a set of knowledge management tools which are the subject of a number of patents. These tools were spun out of BT and are now marketed by Infonic Ltd, of which Dr Davies is Group Technical Advisor. Dr Davies received the BT Award for Technology Entrepreneurship for his contribution to the creation of Infonic.

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Introduction

This book covers a diverse range of important themes in modern information retrieval. We first provide a general overview of information retrieval and then describe the chapters with a roadmap for guidance on the topics covered.

What is Information Retrieval?

Information retrieval, simply put, is about finding information. It encompasses the topics of *information need, search, retrieval*, and *access*. It can involve a range of content and media. Work on the storage and retrieval of data began very soon after the invention of the digital computer. The term *information retrieval* was coined around 1952, and as early as 1945, Bush (1945) was talking about a 'device in which an individual stores all his books, records, and communications, and which is mechanized so that it may be consulted with exceeding speed and flexibility'. The increasing amount and diversity of information that is available to us now, along with the challenges these bring, make this field a continually evolving and exciting area.

Information retrieval (IR) can be explained with the following typical problem (Figure 1): On the one hand, we have a person with an information need. This need, which can initially be vague, somehow needs to be articulated into a request which describes this. The request is then converted into a *search statement* or as is now more often referred to as a *query*. On the other hand, we have information stored in collections. They sit as a potentially valuable resource, but in order to be found they need to be represented somehow and then subsequently indexed. The challenge is to provide a good match between these two in order to ensure the information presented is of relevance to the person with the original query.

More specifically, information retrieval is the process of matching the *query* against the *information objects* that are *indexed*. An index is an optimised data structure that is built on top of the information objects, allowing faster access for the search process. The indexer tokenises the text (parsing), removes words with little semantic value (so-called *stop-words*), and unifies word families (so-called *stemming*). The same is done for the query as well. *Users* express their information need as a *request (search terms)* and it is formulised as a *query* for the retrieval system. The information system responds by *matching information objects*, which are *relevant* to this query. Information retrieval focuses on finding relevant information rather than simple pattern matching. It is also important to note that *relevance* is a subjective notion, since different users may make various judgments about the relevance or non-relevance of particular documents or information objects to given questions.

A *retrieval strategy* (model) is an algorithm and related structures that takes a query and a set of documents and assigns a *similarity measure* between the query and each document. This similarity represents relevance to the user query. Documents are then *ranked* on the basis of their similarity to the query and presented to the user. This process can be repeated and the query can be modified.

Figure 2 depicts the concepts that are usually provided by an information retrieval system.

Information retrieval has a number of models (as will be discussed in Chapter 1). Three main models are the *Boolean model*, the *Vector Space model*, and the *Probabilistic model*. The Boolean model is built on binary relevance (documents are treated either as relevant or non-relevant). It uses *exact match search strategy* based on Boolean expressions (such as AND, OR, NOT). The Vector Space model maps both query and documents in a vector space and uses spatial distance as a similarity measure.

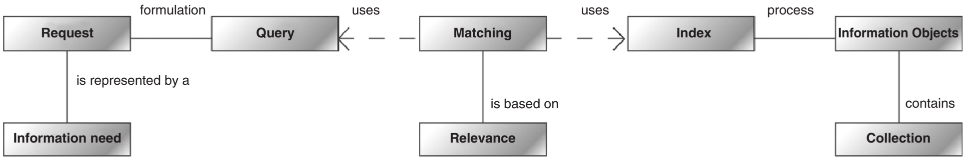

Figure 1 Information retrieval in general

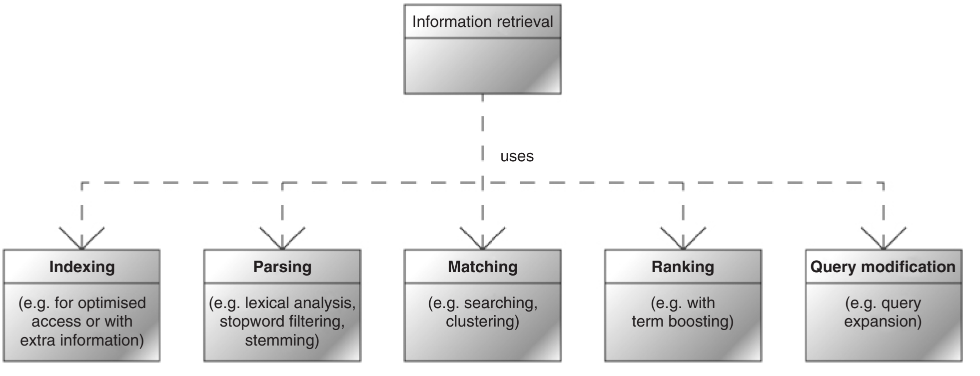

Figure 2 Information retrieval processes

On the other hand, the probabilistic model estimates document relevance as a probability using user feedback for iterative improvement. Both the vector space and the probabilistic model support ranking. There are other models that are also available, but these are a common start to the field.

Information retrieval systems tend to incorporate several activities that are also shown in Figure 2. Broadly speaking, there is *indexing, parsing, matching, ranking*, and *query modification*. These processes are commonly found in search engines. In practice, some of these can be brought together. For example, parsing can be part of an indexing process, and likewise the matching and ranking can be grouped under one umbrella – though not all methods of matching need to have a ranking.

Much of the research and development in information retrieval is aimed at improving retrieval efficiency (see also Chapter 2). Measures referred to as *precision* and *recall* are usually used to measure effectiveness. Precision is the ratio of the number of relevant documents retrieved in relation to the total number of documents retrieved. Recall is the ratio of the number of relevant documents retrieved in relation to the total number of relevant documents. There are other measures too, but these provide a core foundation to the field. If we are to improve information retrieval systems, then it is imperative we have some means of evaluating them against each other.

Overview of the Book

In selecting the chapters for this volume, we have attempted to offer comprehensive coverage of the main themes in modern information retrieval.

We begin with a look at the theoretical underpinnings of the field of information retrieval. As the first chapter explains, there are multiple formal IR models and each may be suited for some IR tasks and less well suited for others, and the chapter offers an excellent introduction to the principal ones in use today. With all IT systems, it is critical to know whether the system performs as intended. Believing therefore that evaluation is an important aspect of information retrieval, and further that the user view of the system and their satisfaction with the retrieved information objects are paramount, we proceed with Chapter 2, on user-centred evaluation.

An increasing proportion of information is non-textual in format: consider, for example, the emergence of *YouTube* and *Flickr* and the many similar systems as web-based sources of video and audio information. With this in mind, the next two chapters are devoted to retrieval of non-textual information resources. In a comprehensive overview of the area, Chapter 3 introduces the basic concepts of multimedia resource discovery, and looks at challenges of multimedia indexing, before discussing novel services which can be offered in the multimedia context. This chapter is followed by a more detailed look at the particular issues raised by image retrieval with a particular focus on user needs analysis (Chapter 4).

Web-based IR has of course emerged as a large area of research and development in its own right over the last 15 years, and the next chapter is devoted to this important topic. The distinctive characteristics of web IR are discussed, and the way in which IR on the web has been developed to deal with these characteristics is described. Like the web, the availability and use of mobile information environments have expanded at a dramatic rate in recent years, and mobile search is the subject of Chapter 6. The diversity and modes of use of mobile information environments globally are discussed and the distinctive features of mobile search identified. Two compelling mobile search applications from very different environments are then described. A key aspect of mobile environments is the requirement for context sensitivity, and a wider discussion on the role of context in IR is the subject of Chapter 7. The importance and role of context information in handling challenges of information search and retrieval are discussed. The capture and representation of context information is described along with examples and guidelines on how a context-aware information system or application can be developed. Consideration is then given to system evaluation in such context-aware information applications.

Unlike the topics covered by some other chapters, categorisation is an area which has been researched since the early days of IR and it continues to be an important topic. A fresh perspective is offered in Chapter 8 by Stuart Watt who interestingly focuses on the purposes of categorisation from the human problem-solving perspective and provides a framework of approaches to further work on categorisation in IR.

Chapter 9 looks at the role of semantic web technology in information retrieval. Central to this approach is the semantic analysis of the target document collection using techniques such as named entity recognition and information extraction, on the hypothesis that this deeper analysis of the target texts may offer improved IR. These semantic analysis techniques are described, followed by a survey of systems using such an approach. Following this chapter, a related topic is covered: the role of natural language technology in IR (Chapter 10). A range of natural language processing (NLP) techniques relevant to IR are described, along with how they can be applied to IR. The usefulness of their contribution to date and future prospects for NLP in IR are discussed. Cross-language IR (CLIR) is an area of increasing interest driven partly by the growing availability of non-English resources (e.g. on the web). Chapter 11 provides a comprehensive overview of the state of the art. We finish with a chapter of central importance to all types of IR: the use of parallel computing to improve the performance of what is by nature a computationally intensive task (Chapter 12). We look at different methods for distributing inverted file data on a parallel computer and the challenges of developing a model by comparing the output of a predictive model against empirical data.

The book is intended for graduate students, later stage undergraduate students, researchers and information/computer scientists, and in particular the growing number of industrial IT professionals whose role requires them to have an understanding of this increasingly important topic. Exercises have been included at the end of each chapter to help readers test and further develop their understanding of the material presented.

Reference

Bush, V. (1945). As we may think. *Atlantic Monthly* **176**(1): 101–108.